PROJECT- ETHICAL PROMPT FILTER using gradio BY ARBON(231030272)

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!pip install gradio nltk spacy
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```

```
import re
import nltk
import spacy
import gradio as gr

nlp = spacy.load("en_core_web_sm")
```

```
# --- ETHICAL PROMPT FILTER ---

harmful_keywords = [
    "hack", "kill", "bomb", "drugs", "doxx", "violence",
    "hate", "racist", "explosive", "malware"
]

def ethical_filter(prompt):
    for word in harmful_keywords:
        if word in prompt.lower():
            return False, f"Unsafe Prompt: contains harmful keyword '{word}'"
```

```
    return True, "Prompt is Safe"
```

```
# --- BIAS DETECTOR ---

protected_groups = [
    "women", "men", "muslim", "hindu", "christian",
    "asian", "black", "white", "lgbt", "trans"
]

stereotype_words = [
    "always", "never", "typically", "all of them",
    "generally", "they are like"
]

def bias_detector(text):
    warnings = []
    lowercase = text.lower()

    for group in protected_groups:
        if group in lowercase:
            for s in stereotype_words:
                if s in lowercase:
                    warnings.append(f"⚠ Possible stereotype about '{group}' using '{s}'")

    return warnings
```

```
# --- HALLUCINATION SCANNER ---

hallucination_phrases = [
    "studies show", "research proves", "according to scientists",
    "89%", "90%", "95%", "everyone knows"
]

def hallucination_scanner(text):
    hall = []

    for phrase in hallucination_phrases:
        if phrase in text.lower():
            hall.append(f"⚠ Possible hallucination: '{phrase}' (no citation)")

    return hall
```

```
# --- COMBINED SYSTEM ---

def analyze_prompt(prompt, model_output):

    # 1. Ethical Filter
    safe, status = ethical_filter(prompt)
    if not safe:
        return status, "", ""

    # 2. Bias Check
    bias = bias_detector(model_output)
    bias_result = "\n".join(bias) if bias else "No Bias Detected"

    # 3. Hallucination Check
    hall = hallucination_scanner(model_output)
    hall_result = "\n".join(hall) if hall else "No Hallucination Detected"

    return status, bias_result, hall_result
```

```
# --- GRADIO INTERFACE ---

with gr.Blocks() as app:
    gr.Markdown("Ethical Prompt Filter + Bias & Hallucination Scanner")

    prompt = gr.Textbox(label="Enter Prompt")
    output = gr.Textbox(label="Enter Model Output (text to analyze)")

    btn = gr.Button("Analyze")

    ethical_res = gr.Textbox(label="Ethical Filter Result")
    bias_res = gr.Textbox(label="Bias Report")
    hallucination_res = gr.Textbox(label="Hallucination Report")
```

```
    btn.click(analyze_prompt, inputs=[prompt, output], outputs=[ethical_res, bias_res, hallucination_res])

app.launch()
```